Training data

$$X_{train} = X = \begin{array}{c} \begin{array}{cccc} \underline{f_1} & \underline{f_2} & \underline{f_3} & \quad\quad \underline{f_{14}} \end{array} \\ \left(\begin{array}{c} 339 \times 14 \\ \\ \\ \end{array} \right) \\ \begin{array}{ccc} \text{CRIM} & \text{ZON} & \text{INDUS} \end{array} \end{array} \qquad \begin{array}{c} \underline{y} \\ \left(\begin{array}{c} \\ \\ \\ \end{array} \right) \end{array} \; 339 \times 1.$$

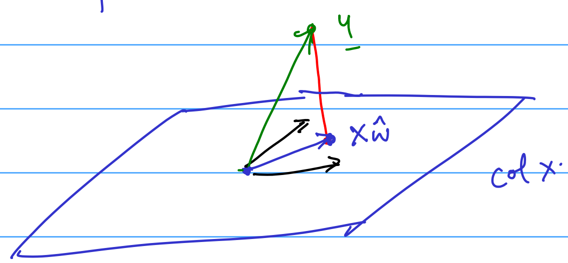
Set of all linear combinations of columns of X : $\text{Col}(X)$.

$X\underline{w}$: If $\underline{w} = \begin{pmatrix} w_1 \\ \vdots \\ w_{14} \end{pmatrix}$ $\quad X\underline{w} = w_1 \underline{f_1} + w_2 \underline{f_2} + \dots + w_{14} \underline{f_{14}}$

Attempt 0 Maybe $\underline{y}$ can be written as $X\underline{w}$?

Solve $\underline{y} = X\underline{w}$. $\quad [X \; \underline{y}]$

Most likely there is no solution.



$\underline{y}$

$X\hat{\underline{w}}$

$\underline{y} - X\hat{\underline{w}} := \underline{e}$

Red line orthogonal to plane: Red line is $\perp$ to every vector in plane.

Suppose $M = \{\underline{v}_1 \dots \underline{v}_n\}$ is a maximal linearly indpt set in $\text{Col}(X)$.

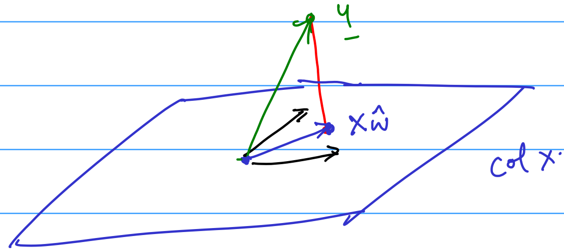
(What would be a maximal lin. ind. set in $\text{col}(X)$?)

Suppose. $\underline{e}^T \underline{v}_1 = \underline{e}^T \underline{v}_2 \dots \underline{e}^T \underline{v}_n = 0.$

Pick $\underline{x} \in \text{col}(X)$. Now $[\underline{v}_1 \dots \underline{v}_n \; \underline{x}]$ has the last col free.

$\therefore \underline{x} = \begin{bmatrix} \underline{v}_1 & \dots & \underline{v}_n \end{bmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}.$

$$\underline{e}^T \underline{x} = \underline{e}^T \begin{bmatrix} \underline{v}_1 & \dots & \underline{v}_n \end{bmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} = \begin{bmatrix} \underline{e}^T \underline{v}_1 & \dots & \underline{e}^T \underline{v}_n \end{bmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} = 0.$$



$\underline{y}$

$X\underline{\hat{w}}$

$y - X\underline{\hat{w}} = \underline{e}$

What is the closest point to y? It's $X\hat{w}$ such that $y - X\underline{\hat{w}}$ is orthogonal to every pivot col of X.

Assuming all cols of X are pivot cols. $y - X\underline{\hat{w}} \perp$ every col of X.

$$X = \begin{bmatrix} | & & | \\ \underline{f}_1 & \cdots & \underline{f}_{14} \\ | & & | \end{bmatrix} \qquad X^T = \begin{bmatrix} — & \underline{f}_1^T & — \\ & \vdots & \\ — & \underline{f}_{14}^T & — \end{bmatrix}$$

$$X^T \underline{e} = \begin{bmatrix} — & \underline{f}_1^T & — \\ & \vdots & \\ — & \underline{f}_{14}^T & — \end{bmatrix} \underline{e} = \begin{bmatrix} \underline{f}_1^T \underline{e} \\ \vdots \\ \underline{f}_{14}^T \underline{e} \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} = \underline{0}.$$

$X^T \underline{e} = \underline{0}$ is the formalization that $\underline{e}$ is $\perp$ to all vectors in col(X).

$$\underline{0} = X^T \underline{e} = X^T (\underline{y} - X\underline{\hat{w}}).$$
$$= X^T \underline{y} - X^T (X\underline{\hat{w}})$$
$$= X^T \underline{y} - (X^T X)\underline{\hat{w}}$$

$X : n \times p$. $\quad X^T : p \times n$

$\underset{p \times n}{X^T}\,\underset{n \times p}{X} : p \times p$. $\qquad (X^T X)^T = X^T (X^T)^T = \underline{X^T X}$

$$(AB)^T = B^T A^T$$

$$\left[\begin{pmatrix} | & | \\ \underline{a}_1 & \underline{a}_2 \\ | & | \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}\right]^T = \left[b_1 \underline{a}_1 + b_2 \underline{a}_2\right]^T = b_1 \underline{a}_1^T + b_2 \underline{a}_2^T$$
$$= (b_1 \quad b_2) \begin{bmatrix} — & \underline{a}_1^T & — \\ — & \underline{a}_2^T & — \end{bmatrix}.$$
$$= \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}^T A^T.$$

$$\begin{pmatrix} a & c \\ d & b \end{pmatrix}$$

Transpose

$$\begin{pmatrix} a & d \\ c & b \end{pmatrix}$$